

```
In [1]: import nltk
        from nltk.tokenize import WhitespaceTokenizer, WordPunctTokenizer, TreebankWordTokenizer
        from nltk.stem import PorterStemmer, SnowballStemmer
        from nltk.stem import WordNetLemmatizer
```

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In [2]: nltk.download('punkt')
        nltk.download('wordnet')
        nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data]   C:\Users\Admin\AppData\Roaming\nltk_data...
```

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Out[2]: True
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In [3]: text = "NLTK's tokenizers are great! Let's test whitespace, punctuation, treebank, tweet, & MWE tokenization."
```

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In [4]: ws_tokens = WhitespaceTokenizer().tokenize(text)
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In [5]: wp_tokens = WordPunctTokenizer().tokenize(text)
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In [6]: tb_tokens = TreebankWordTokenizer().tokenize(text)
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In [7]: tweet_tokens = TweetTokenizer().tokenize(text)
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In [8]: mwe_tokenizer = MWETokenizer([('New', 'York'), ('machine', 'learning')])
        mwe_tokens = mwe_tokenizer.tokenize(tb_tokens)
        print("Whitespace:", ws_tokens)
        print("Punctuation:", wp_tokens)
        print("Treebank:", tb_tokens)
        print("Tweet:", tweet_tokens)
        print("MWE:", mwe_tokens)
```

```
Whitespace: ["NLTK's", 'tokenizers', 'are', 'great!', "Let's", 'test', 'whitespace,',
'punctuation,', 'treebank,', 'tweet', '&', 'MWE', 'tokenization.']
Punctuation: ['NLTK', "'", 's', 'tokenizers', 'are', 'great', '!', 'Let', "'", 's',
'test', 'whitespace', ',', 'punctuation', ',', 'treebank', ',', 'tweet', '&', 'MWE',
'tokenization', '.']
Treebank: ['NLTK', "'", 's', 'tokenizers', 'are', 'great', '!', 'Let', "'", 's', 'test', 'whi
tespace', ',', 'punctuation', ',', 'treebank', ',', 'tweet', '&', 'MWE', 'tokenization', '.']
Tweet: ["NLTK's", 'tokenizers', 'are', 'great', '!', "Let's", 'test', 'whitespace',
',', 'punctuation', ',', 'treebank', ',', 'tweet', '&', 'MWE', 'tokenization', '.']
MWE: ['NLTK', "'", 's', 'tokenizers', 'are', 'great', '!', 'Let', "'", 's', 'test', 'whitespa
ce', ',', 'punctuation', ',', 'treebank', ',', 'tweet', '&', 'MWE', 'tokenization',
'.']
```

```
In [9]: porter = PorterStemmer()
        snowball = SnowballStemmer("english")
        porter_stems = [porter.stem(token) for token in tb_tokens]
        snowball_stems = [snowball.stem(token) for token in tb_tokens]
```

```
print("Porter Stemmer:", porter_stems)
print("Snowball Stemmer:", snowball_stems)
```

```
Porter Stemmer: ['nltk', "'s", 'token', 'are', 'great', '!', 'let', "'s", 'test', 'whitespace', ',', 'punctuat', ',', 'treebank', ',', 'tweet', '&', 'mwe', 'token', '.']
Snowball Stemmer: ['nltk', "'s", 'token', 'are', 'great', '!', 'let', "'s", 'test', 'whitespace', ',', 'punctuat', ',', 'treebank', ',', 'tweet', '&', 'mwe', 'token', '.']
```

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In [10]: lemmatizer = WordNetLemmatizer()
lemmas = [lemmatizer.lemmatize(token.lower()) for token in tb_tokens]
print("Lemmatized:", lemmas)
```

```
Lemmatized: ['nltk', "'s", 'tokenizers', 'are', 'great', '!', 'let', "'s", 'test', 'whitespace', ',', 'punctuation', ',', 'treebank', ',', 'tweet', '&', 'mwe', 'tokenization', '.']
```

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In [ ]:
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